

Executive Summary

Serious games (SGs) for education increasingly integrate adaptive assessment and analytics to personalize learning. Recent work applies **fuzzy logic** to model uncertain student states and adapt game content. For example, Papadimitriou et al. (2019) developed *FuzzEG*, an HTML-adventure SG using fuzzy sets to represent learner knowledge levels and dynamically adjust quiz difficulty; a study with 60 students showed high usability and effective adaptivity ¹. Similarly, Chrysafiadi et al. (2022) created a fuzzy student model to adapt both content difficulty and game narrative in an educational adventure game, yielding high acceptance by learners and teachers ² ³. These works exemplify a trend: **fuzzy-based personalization** of SGs can improve engagement and learning outcomes by tailoring challenges in real time.

In parallel, **learning analytics and game telemetry** have matured as tools for SG evaluation and assessment. Lu et al. (2023) engineered a telemetry system for a science SG, capturing 60 in-game features from 373 students; they used data mining to build predictive models of student performance, identifying key gameplay events linked to learning gains ⁴. In surveys of game difficulty adaptation (Paraschos & Koulouriotis 2022) and SG analytics (Alonso-Fernández et al. 2019), researchers report that collecting rich game logs and applying machine learning enables fine-grained, **adaptive assessment** of learners, far beyond traditional pre/post tests.

Finally, **evaluation frameworks and reviews** have codified best practices. Systematic reviews (e.g., Calderón & Ruiz 2015) catalog evaluation methods in SGs ⁵, and Wilson et al. (2016) propose that SG evaluation cover four “groundings” (technical, empirical, theoretical, external). Modern frameworks even incorporate digital twins and semantic metadata to adjust SGs in real time ⁶. Across this literature, key themes are personalization (especially via fuzzy adaptation), data-driven assessment (telemetry analytics), and rigorous evaluation frameworks (theory + empirical validation).

Research Gaps: Despite these advances, gaps remain. (1) Many SGs adapt content or difficulty separately, but **integrated adaptive assessment** combining fuzzy student modeling with real-time analytics is rare. (2) While telemetry studies capture rich data, the challenge of **linking in-game metrics to learning outcomes under uncertainty** needs more work (e.g. integrating fuzzy reasoning with machine learning). (3) Few studies examine **scalability and generalizability**: most user evaluations are small. Our PFE can address these by developing a fuzzy-logic evaluation system that ingests SG telemetry to produce multi-criteria adaptive feedback, and testing it in a realistic deployment.

Annotated Bibliography

Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
Papadimitriou, S., Chrysafiadi, K., & Virvou, M. (2019). <i>FuzzEG: Fuzzy logic for adaptive scenarios in an educational adventure game</i>. Multimedia Tools and Applications, 78(22), 32023–32053.	Presents “FuzzEG”, an HTML-programming adventure game that uses fuzzy sets to represent a learner’s knowledge level and adapt quiz content and difficulty in real time ¹. The system categorizes knowledge into fuzzy groups (e.g. “Unknown” to “Learned”) and applies rules to expand or contract the scenario based on progress ¹. In an evaluation with 60 postgraduate students, users reported high usability, educational effectiveness, and satisfaction with the adaptive gameplay ⁷. This work demonstrates fuzzy logic’s value for personalizing serious games.	<i>fuzzy logic; adaptive learning; educational games; knowledge modeling; user evaluation</i>	2019, Springer (MTAP)	High	10.1007/s11042-019-07955-w

Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
Papadimitriou, S., Chrysafiadi, K., & Virvou, M. (2019). <i>Evaluating the use of fuzzy logic in an educational game for offering adaptation</i>. Proc. IEEE CITS (pp. 1–5).	Conference study applying fuzzy logic to adapt an educational game. (Summarized in DBLP ⁸.) It describes a prototype game that tailors content based on fuzzy assessment of player skill, then reports preliminary usage. The short paper confirms feasibility of on-the-fly fuzzy adaptation in SGs, though details are limited due to conference length.	<i>fuzzy logic; educational game; adaptation; prototype evaluation</i>	2019, IEEE CITS	Medium	DOI: 10.1109/CITS.2019.8862064

Papadimitriou, S., Chrysafiadi, K., & Virvou, M. (2023). <i>Fuzzy-based dynamic difficulty adjustment of an educational 3D-game</i>. Multimedia Tools and Applications, 82, 27525–27549.	Describes a 3D HTML-learning game that uses fuzzy logic to adjust game difficulty dynamically. The system tracks two skill dimensions (combat and maze navigation) using fuzzy rules on metrics like time taken, player damage, and help requests ⁹. It then increases/decreases enemy count, maze traps, etc., in real time ¹⁰. A study with 34 undergraduates found the fuzzy-driven DDA kept players engaged and improved motivation (fewer dropouts) compared to static difficulty ¹¹. Unlike data-hungry AI methods, this approach adapts on-the-fly without extensive training data ¹². This recent	<i>dynamic difficulty; adaptive gameplay; fuzzy rules; educational game; user study</i>	2023, Springer (MTAP)	High	10.1007/s11042-023-14515-w
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	work highlights fuzzy DDA's practical impact in SGs.				

Chrysafiadi, K., Papadimitriou, S., & Virvou, M. (2022). <i>Cognitive-based adaptive scenarios in educational games using fuzzy reasoning. Knowledge-Based Systems</i>, 250, 109111.	Implements an educational adventure game that adapts both content difficulty and story narrative using a fuzzy student model. The model infers the learner's cognitive state (knowledge/ weaknesses) and decides whether to extend the game scenario or choose harder exercises ² . Adaptation is performed at each step via fuzzy logic rules. In evaluation, learners and teachers reported high acceptance, and learning performance improved when the plot was adapted to individual needs ³ . This study is highly relevant: it shows <i>fuzzy reasoning</i> can personalize multiple dimensions of	<i>fuzzy student model; adaptive scenarios; personalized learning; serious games; cognitive state</i>	2022, Elsevier (KBS)	High	10.1016/j.knosys.2022.109111
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	an SG (not just difficulty), reinforcing fuzzy logic's role in adaptive assessment.				

Lu, W., Santos, J. L., & Keller, J. (2023). <i>Serious game analytics by design: Feature generation and selection using game telemetry and game metrics – Toward predictive model construction</i>. Journal of Learning Analytics, 10(3), 154–178.	Develops an end-to-end analytics framework for a serious game ("Mission HydroSci") teaching water science. The authors instrumented the game to log 60 features of player actions (e.g. quiz answers, time usage) for 373 students. They applied feature selection to identify key predictors of learning outcomes, built a predictive model of post-test scores, and provided a dashboard for instructors ⁴. The study found that a small subset of game metrics (e.g. correct actions, decision paths) strongly correlated with learning gains ⁴. This work is relevant as it shows how telemetry can produce <i>quantitative assessments</i> of	<i>learning analytics; game telemetry; predictive modeling; educational games; performance metrics</i>	2023, Society for Learning Analytics (JLA)	High	10.18608/jla.2023.1054
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	SG learning, enabling automatic evaluation. The paper also outlines technical methods (log schema, analytics pipeline) useful for any SG assessment system.				

Paraschos, P. D., & Koulouriotis, D. E. (2022). <i>Game difficulty adaptation and experience personalization: A literature review</i>. International Journal of Human-Computer Interaction, 38(3), 189–211.	A systematic review (2005–2021) of adaptive and personalized gaming. The authors survey 109 works on dynamic difficulty adjustment, procedural content generation, and user modeling. They categorize approaches by adaptation type (e.g. difficulty, content, UI) and underlying methods (AI planning, affect modeling, etc.). Notably, the review finds that adaptive/ personalized SGs tend to improve player engagement and learning, often by modeling individual traits and skills ¹³. While not limited to education, this paper identifies fuzzy logic as one viable DDA technique among others. It provides broad context on	<i>dynamic difficulty; personalization; user modeling; procedural content; adaptive games</i>	2022, Taylor & Francis (IJHCI)	Medium	10.1080/10447318.2021.2020008
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	personalization trends in SGs and can inspire hybrid methods.				

Djaouti, D., Dau & al. (2024). <i>A Scoping Review on Fuzzy Logic Used in Serious Games.</i> Technologies, 13(10), 448.	Comprehensive scoping review of fuzzy logic in SGs (2020– 2025). It maps 84 studies, reporting that ~80% target educational SGs (the rest health) ¹⁴ . The review details how fuzzy inference systems are implemented (Mamdani-type models, membership functions, rules) and in what game genres and contexts. It finds fuzzy logic is predominantly used for adaptive features (e.g. difficulty adjustment, feedback) in 2D/3D games. The paper’s taxonomy (application domains, interaction modes, etc.) and analysis of experiment protocols (participants, metrics, statistical tests) provide a valuable reference on	<i>fuzzy logic; serious games; systematic review; taxonomy; adaptive systems</i>	2024, MDPI (Technologies)	High	10.3390/technologies13100384
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	the state of fuzzy SG research ¹⁵ ¹⁴ . Very high priority: it synthesizes recent trends in adaptive SGs and highlights gaps (e.g. few non-visual feedback uses).				

Calderón, A., & Ruiz, M. (2015). <i>A systematic literature review on serious games evaluation: An application to software project management. Computers & Education</i> , 87, 396–422.	This systematic review (2015) examines how researchers evaluate SGs. From 1199 initial hits, 102 primary studies were selected. The authors categorize the evaluation methods (surveys, tests, telemetry analysis, etc.), game domains, and learning outcomes assessed ⁵ . They report common features evaluated (knowledge gain, engagement, usability) and typical participant sample sizes. Although focused on project-management SGs, the review's taxonomy of evaluation procedures is generalizable. Its findings (e.g. over-reliance on pre/post tests, small cohorts) highlight evaluation best	serious games evaluation; assessment methods; systematic review; educational effectiveness	2015, Elsevier (C&E)	Medium	10.1016/j.compedu.2015.07.011
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	practices and weaknesses. This reference is medium priority: it outlines established SG evaluation methods, against which adaptive/fuzzy evaluation innovations can be compared				
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Al-Qaimari, H. H., & Anjum, A. (2023). A framework for standardizing the development of serious games with real-time self-adaptation capabilities using digital twins. Applied Sciences, 13(8), 369.	Proposes a general framework for building adaptive SGs using <i>digital twins</i> and semantic annotation. Games' elements and data are modeled with ontologies to enable self-adaptation: a digital twin monitors player performance and applies expert-defined rules to adjust difficulty, content, etc. in real time ⁶ ¹⁶. Two case studies (child therapy game and industrial training) show improved engagement via on-the-fly personalization. This work is medium priority: it contributes a structured development process (using ontologies and DTs) for adaptive SGs, which could inform design of our	<i>digital twin; semantic modeling; self-adaptation; serious game development</i>	2023, MDPI (Applied Sci.)	Medium	10.3390/app13083690
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	evaluation system, though it does not specifically use fuzzy logic.				

Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
Wilson, D. W., Jenkins, J. L., & Nunamaker, J. F. Jr. (2016). <i>Serious games: An evaluation framework and case study</i> . In Proc. 49th Hawaii Int. Conf. System Sciences .	Introduces a four-part evaluation framework for SGs: theoretical, technical, empirical , and external grounding ¹⁷ . The model emphasizes validating educational design (theoretical), software functionality (technical), empirical learning outcomes (empirical), and broader context (external). The paper reports a case study applying this framework to a language-learning SG. It's medium priority: it offers a broad structure for comprehensive SG evaluation that could guide our assessment approach, though it predates fuzzy and analytics methods.	<i>evaluation framework; serious games design; theoretical grounding; empirical assessment</i>	2016, IEEE (HICSS)	Medium	[Reference unavailable online]

Alonso-Fernández, C., Calvo-Morata, A., Freire, M., Martínez-Ortiz, I., & Fernández-Manjón, B. (2019). <i>Applications of data science to game learning analytics data: A systematic literature review</i>. Computers & Education, 141, 103612.	Systematic review of how data science (machine learning, statistical analysis) is applied to game-learning data. The authors classify studies by objectives (e.g. prediction, adaptation), data sources, and analytic techniques. They report that most work focuses on modeling learner behavior and personalization using techniques like clustering and regression. The review underscores the need for standardized GLA frameworks and shared datasets. High priority: it directly surveys analytical methods applied to SG telemetry, aligning with our goal of predictive evaluation	<i>game learning analytics; data science; systematic review; learner modeling; predictive analytics</i>	2019, Elsevier (C&E)	High	10.1016/j.compedu.2019.103612
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Daoudi, I., Dendani, Y., Meziane, F., & Ennaji, H. (2022). <i>Learning analytics for enhancing the usability of serious games in formal education: A systematic literature review and research agenda</i> . IEEE Access , 10, 103963–103983.	This systematic review (2022) collected 80 studies on applying learning analytics (LA) in educational SGs. It summarizes core LA methods (e.g. xAPI, data mining, dashboards) and identifies design features of LA-enabled SGs. For instance, earlier work like Malliarakis et al. (2014) proposed a conceptual framework combining profiling and adaptation in an MMORPG 19 . High priority: the paper highlights that most SG-LA efforts focus on behavioral analytics and suggests open challenges in LA integration. It provides a research agenda for using student interaction data to improve	<i>learning analytics; serious games; systematic review; xAPI; educational data mining</i>	2022, IEEE (Access)	High	10.1109/ACCESS.2022.3160554
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	game design and assessment in education.				

Primanita, A., Satria, H., Rizqie, M. Q., Iskandar, A. H., & Nugraha, W. (2025). <i>Adaptive hint generation for educational games using fuzzy logic</i>. Jurnal Teknik Informatika, 18(1), 101–110.	Proposes a fuzzy-logic system to generate in-game hints for players struggling with programming puzzles. The fuzzy model takes as input a learner's success rate and frustration level to output contextual hints. In simulations and a classroom trial (using the GUESS-18 framework), the system delivered more hints to beginner players and scored high on playability/ usability ²⁰ . This shows fuzzy logic can be used not just for difficulty adjustment but for <i>formative assistance</i> in SGs. Medium priority: it offers a novel use of fuzzy logic for adaptive scaffolding, which could inform hint/ feedback	<i>fuzzy logic; hint generation; programming game; adaptive assistance; usability</i>	2025, Indonesia (Jurnal TI)	Medium	10.15408/jti.v18i1.41893
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Citation	Annotation	Keywords	Year/Venue	Priority	Link/DOI
	design in our evaluative SG.				

Synthesis and Research Gaps

These sources reveal that **adaptive evaluation in serious games** is an active, multidisciplinary field. Key trends include the use of fuzzy logic to handle uncertain aspects of learner models (e.g. knowledge proficiency, engagement) ¹ ² and the integration of telemetry/learning analytics to quantify player actions ⁴ ¹⁸ . Many works (Papadimitriou 2019, Chrysafiadi 2022, Primanita 2025) apply fuzzy reasoning for real-time adaptation, often showing improved engagement in user studies ⁷ ¹¹ ³ . Simultaneously, reviews (Alonso-Fernández 2019; Daoudi 2022) emphasize data-driven methods to analyze game data and adapt experiences. Together, the literature suggests **personalization** (fuzzy or data-driven) can make SGs more effective.

However, three research gaps emerge. **First**, there is no unified system combining fuzzy-logic student models with dynamic analytics for *automated adaptive assessment* in SGs; most studies treat these separately. **Second**, linking in-game behavioral metrics to learning outcomes under uncertainty remains underexplored. While Lu et al. (2023) built predictive models from features ⁴ , few use fuzzy inference to interpret those metrics. **Third**, existing evaluations are often small-scale (tens of users) and domain-specific. A gap exists in *scalable evaluation frameworks* that leverage fuzzy/LA techniques across diverse educational games.

Our PFE can address these gaps by developing a holistic framework that uses fuzzy logic rules over gameplay data to evaluate learner performance continuously, and adapting feedback or game content accordingly. This could combine the personalization strengths of fuzzy models (Papadimitriou 2019; Chrysafiadi 2022) with the predictive power of game analytics (Lu 2023).

Gap-to-Contribution Mapping

Identified Gap	Proposed PFE Contribution
1. Integrated adaptive assessment: Few SGs systematically integrate fuzzy student models with analytics to automatically assess and adapt learning in real time.	Fuzzy-analytics evaluation system: We will design a framework that ingests SG telemetry and applies a fuzzy inference engine (modeling knowledge/confidence levels) to compute adaptive assessment scores and adjust in-game challenges or hints on the fly.
2. Uncertain metric interpretation: In-game metrics are often treated as crisp variables. There is limited use of fuzzy reasoning to interpret continuous or subjective game data (e.g. hint usage, decision times).	Fuzzy interpretation of metrics: Our system will fuzzify raw game telemetry (e.g. time to answer, health lost) into linguistic states (e.g. “slow”, “few lives lost”) and use rule sets to infer latent skills or needs, enabling nuanced feedback beyond binary correct/incorrect.

Identified Gap	Proposed PFE Contribution
3. Scalable evaluation framework: Most SG evaluations use small samples and lack a general methodology for adaptive games.	Generalizable evaluation pipeline: We will develop an evaluation pipeline (including a telemetry logger, fuzzy evaluation model, and expert-informed rules) that can be applied to any educational SG. Through a case study in our project domain, we will demonstrate its scalability and adaptability to different content.

Recommended Search Strategies

Search queries: Use combinations like

- "fuzzy logic" "serious games" adaptive learning evaluation
- "serious games" "learning analytics" assessment"
- "game telemetry" evaluation "education"
- "dynamic difficulty adjustment" educational game review
- "serious games" "fuzzy inference" adaptive

Databases: Prioritize scholarly databases and publishers, e.g.

- **IEEE Xplore, ACM Digital Library** (conference/journal papers on educational technology and game AI)
- **SpringerLink, ScienceDirect** (journals like *Computers & Education, Knowledge-Based Systems*)
- **Google Scholar / Semantic Scholar** (to find interdisciplinary and newer publications)
- **MDPI, IEEE Access** (open-access sources)

Continue the literature search iteratively using keywords above, including French works where relevant (e.g. supervised theses on jeux sérieux), and checking references of key papers.

1 7 (PDF) FuzzEG: Fuzzy logic for adaptive scenarios in an educational adventure game

https://www.academia.edu/80911984/FuzzEG_Fuzzy_logic_for_adaptive_scenarios_in_an_educational_adventure_game

2 3 Cognitive-based adaptive scenarios in educational games using fuzzy reasoning - ScienceDirect

<https://www.sciencedirect.com/science/article/abs/pii/S0950705122005482>

4 (PDF) Serious Game Analytics by Design: Feature Generation and Selection Using Game Telemetry and Game Metrics: Toward Model Construction

https://www.researchgate.net/publication/369027123_Serious_Game_Analytics_by_Design_Feature_Generation_and_Selection_Using_Game_Telemetry_and_Game_Metrics_Toward_Predictive_Model_Construction

5 A systematic literature review on serious games evaluation: An application to software project management - ScienceDirect

<https://www.sciencedirect.com/science/article/abs/pii/S0360131515300166>

6 16 A Framework for Standardizing the Development of Serious Games with Real-Time Self-Adaptation Capabilities Using Digital Twins

<https://www.mdpi.com/2227-7080/13/8/369>

8 dblp: Spyros Papadimitriou

<https://dblp.org/pid/139/2998.html>

9 10 11 12 Fuzzy-based dynamic difficulty adjustment of an educational 3D-game | Multimedia Tools and Applications | Springer Nature Link

<https://link.springer.com/article/10.1007/s11042-023-14515-w>

13 Game Difficulty Adaptation and Experience Personalization: A Literature Review

<https://ouci.dntb.gov.ua/en/works/7pbWNabl/>

14 15 A Scoping Review on Fuzzy Logic Used in Serious Games | MDPI

<https://www.mdpi.com/2227-7080/13/10/448>

17 Serious Games: An Evaluation Framework and Case Study | Request PDF

https://www.researchgate.net/publication/300410461_Serious_Games_An_Evaluation_Framework_and_Case_Study

18 19 Learning analytics for enhancing the usability of serious games in formal education: A systematic literature review and research agenda - PMC

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9061227/>

20 (PDF) Adaptive Hint Generation for Educational Games Using Fuzzy Logic

https://www.researchgate.net/publication/391416308_Adaptive_Hint_Generation_for_Educational_Games_Using_Fuzzy_Logic